Peter Keel

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EXPERIENCE

DRW *Chicago, IL*
Market Data Systems Engineer Apr 2024 – Present

- Built a continuous futures roll system used daily by a trading desk to determine when and how to roll positions across 190+ products and 36 exchanges
- Curate historical market data relied on by 300+ quants, traders, and researchers: symbology mapping, daily quality checks, and Parquet conversion so teams get clean, analysis-ready data
- Drove a firm-wide migration to versioned research databases after analyzing a month of query logs across 100+ users to decide what to keep, merge, or retire
- Built an options data layer across CME, ICE, and Eurex, handling quirks like CME's scaled-integer strikes, with a warm cache that returns the latest snapshot of 300,000 symbols in ~1s; built dividend and corporate actions pipelines for pricing models
- Added point-in-time query support to the firm's time-series library; helped build an MCP server with custom skills that lets AI assistants query internal tick data in OneTick
- Own one of the firm's market data platforms day-to-day; run onboarding demos for trading, risk, and research teams

Cboe Global Markets *Chicago, IL*
Product Manager Nov 2021 – Apr 2024

- Supported options data products (Cboe DataShop, LiveVol, Trade Alert): OPRA-based feeds, historical data, implied vols, Greeks, and theoretical values used in obvious-error reviews, delivered via APIs and user-facing tools
- Took an automated NewsFeed from spec to production, running sprints with developers and building API docs in Python and Jinja
- Automated quarterly index rebalance validation in Python, cutting hours of manual checks each cycle
- Managed an intern end-to-end, from scoping the project to the final presentation to group leaders

Nasdaq *Rockville, MD*
Senior Index Analyst Apr 2021 – Nov 2021
Index Analyst Apr 2020 – Apr 2021

- Rewrote 20+ Access/Excel index models in Python, making them 3x faster and cutting reconstitution time 30–80%; presented results and trained the team
- Set up daily automated checks that flag discrepancies in index levels, constituent changes, and corporate actions before they reach clients
- Mapped Bloomberg fixed income symbology to ICE during a vendor migration; reconciled identifier mismatches across thousands of bonds

Wells Fargo *Minneapolis, MN*
Data Science Intern Dec 2019 – Jan 2020

- Built a Python web app routing portfolio managers' bond orders to traders, prioritized by in-house credit ratings

Piper Jaffray Companies *Minneapolis, MN*
Analytics Fixed Income Intern Jun 2019 – Dec 2019

- Developed statistical models to assign credit scores to municipalities, predict corporate bond rating changes, and predict defaults using data cleaning, parameter tuning, and train/test validation

PROJECTS

U.S. Housing Market Explorer *2026*

- Built an interactive geospatial dashboard visualizing housing data across 894 U.S. metros from 8 data sources (Zillow, Census ACS, FRED, BEA, FEMA, EPA), with weekly data refresh automated via GitHub Actions · Live demo

NBA Hall of Fame Prediction Model *2020 / 2026*

- Modeled NBA Hall of Fame induction probability for 380+ players using random forest with 10-fold cross-validation (94.7% accuracy); built in R/Shiny (2020), rebuilt in Python/Streamlit (2026) · Live demo

SKILLS

Coding: Python (Pandas, Scikit-learn, SciPy, Plotly), SQL, R, Bash, JavaScript, HTML/CSS
Data & Tools: OneTick, Bloomberg, Parquet, Trino, Databricks, AWS, Terraform, Docker, Airflow, Git, Linux

EDUCATION

Carleton College *Northfield, MN*
Bachelor of Arts in Mathematics, GPA: 3.64 March 2020

- **Relevant Coursework:** Real Analysis, Multivariable Calculus, Linear Algebra, Probability, Statistical Inference, Applied Regression, Bayesian Statistics, Statistical Consulting, Combinatorics, Data Science, Data Structures, Software Design, Programming Languages, Microeconomics, Macroeconomics
- **Extracurricular:** Varsity Soccer & Track (Academic All-Conference), Carleton Investment Office Intern, Math Tutor

Dartmouth Tuck School of Business *Hanover, NH*
Bridge Program Dec 2018